

LLM-Based Classification of Clinical Complaints

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Background & Motivation

Clinical complaint text is messy, but downstream systems need standardized labels.

CLINICAL PROBLEM

- Patient complaints are often short, informal, and variable in wording.
- The same underlying complaint can be written or spoken in several different ways.
- Manual standardization is possible, but it does not scale well for large datasets or automated workflows.

MODELING OPPORTUNITY

- LLMs can interpret flexible natural language without requiring a hand-written rule for every phrase.
- A closed-set label list keeps the task controlled and evaluable.
- Confidence scores can be used to route uncertain or high-risk cases to human review.

Key framing: this is a supervised decision-support workflow for complaint organization, not an autonomous diagnostic tool.

Project Goal & Questions

Build and evaluate a controlled LLM workflow for complaint labeling.

PROJECT GOAL

Compare zero-shot and few-shot LLM prompting for closed-set clinical complaint classification on both text and audio-derived transcripts.

CORE QUESTIONS

- Does adding retrieved labeled examples improve classification accuracy?
- Does performance change when the transcript comes from automatic speech recognition?
- Which labels are most frequently confused?
- Can confidence thresholds reduce review workload while preserving safety?
- Which audio-quality subgroups deserve additional stress testing?

Dataset Overview

The workflow uses fixed train/validate/test splits and a 25-label closed set.

75

development examples
validate, gold transcripts

500

final gold examples
test, reference transcripts

500

final ASR examples
test, audio → transcript

25

allowed complaint labels
~20/class final stages

EXPERIMENTAL SETUP

- Closed-set classification: each prediction must choose one allowed clinical complaint label.
- Few-shot retrieval pool comes only from the training split, which prevents test-label leakage.
- Development stage uses validation examples and gold transcripts for workflow sanity checks.
- Final gold isolates classifier behavior using reference transcripts.
- Final ASR tests the full audio-to-label pathway using Whisper large-v3-turbo transcripts.

Workflow Completion & Data Quality

The submitted workflow produced complete final-stage artifacts with no recorded prediction errors.

RUN STATUS

Stage	Status	Examples	Errors
Development	Completed	75	0
Final Gold	Completed	500	0
Final ASR	Completed	500	0

ARTIFACT SET

- results.csv / results.json for row-level predictions
- summary.json for stage-level metrics
- confusion matrices and per-class metrics
- subgroup metrics for audio-quality effects
- review economics and calibration figures

Submission-clean wording: this run had zero recorded prediction errors. Fallback handling still belongs in future deployment plans, but not as a reported result.

ML Pipeline & Methodology

The workflow separates clean-transcript classification from the full audio-to-label pipeline.



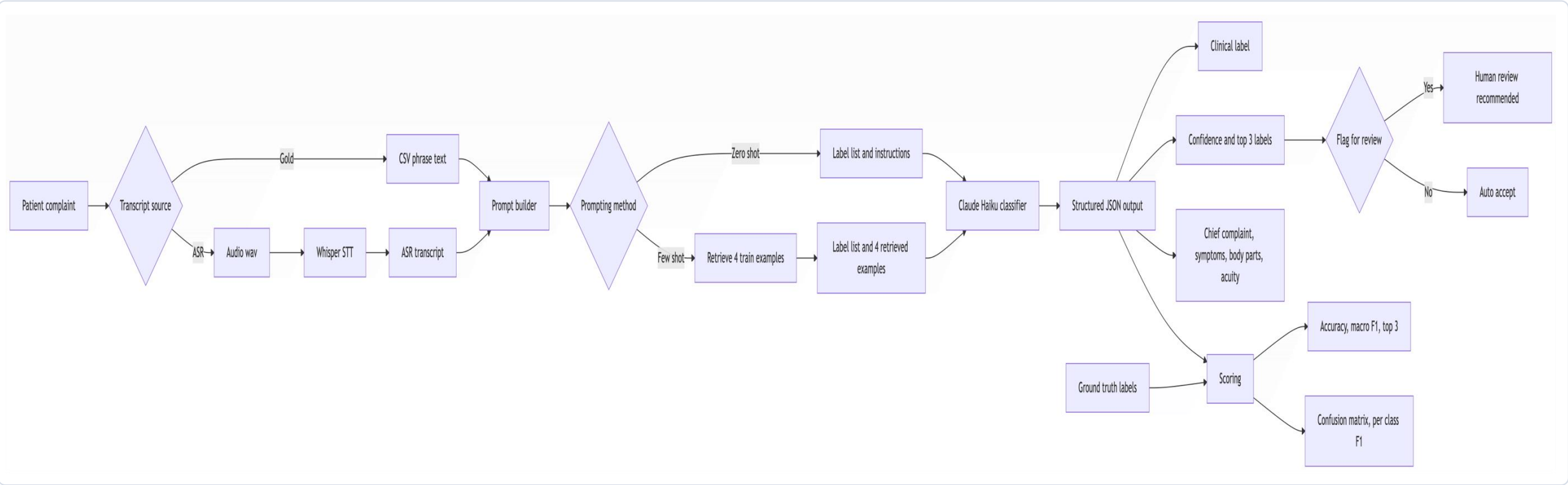
KEY TECHNICAL CHOICES

- Gold path: transcript → classifier → review policy → metrics.
- ASR path: audio → speech-to-text → classifier → review policy → metrics.
- Bootstrap resampling uses 300 samples to estimate 95% confidence intervals.
- Structured response fields include `clinical_label`, `confidence`, `top_3_labels`, `chief_complaint`, `symptoms`, `body_parts`, and `acuity`.

End-to-End Workflow Diagram

This view shows how transcript source, prompting strategy, structured output, review routing, and scoring connect in one workflow.

WORKFLOW DIAGRAM



How to read the diagram: the workflow first branches by transcript source (gold vs ASR), then by prompting method (zero-shot vs few-shot), then sends the structured classifier output both to human-review routing and to scoring against ground-truth labels.

Modeling Methods: Zero-Shot vs Few-Shot

Few-shot retrieval gives the LLM local examples without retraining the model.

ZERO-SHOT

- Transcript
- Allowed 25-label list
- Prompt instructions
- No examples shown
- Simpler and slightly cheaper

FEW-SHOT RETRIEVAL

- Transcript + same label list
- Prompt instructions
- Retrieves $k = 4$ similar training examples
- Examples selected by lexical similarity
- More context for ambiguous wording

Important wording: few-shot retrieval does not retrain the LLM. It conditions the model by showing similar labeled examples at inference time.

Evaluation Metrics & Review Policy

The evaluation measures both classification performance and operational safety.

PERFORMANCE METRICS

- Accuracy: percentage of examples where the top predicted label exactly matches the true label.
- Macro-F1: class-balanced score, useful because it prevents strong labels from hiding weak labels.
- Top-3 accuracy: whether the correct label appears anywhere in the model's top three choices.
- Bootstrap 95% CIs: uncertainty estimated from 300 resampled test sets.

REVIEW POLICY

0.60

standard confidence threshold

0.80

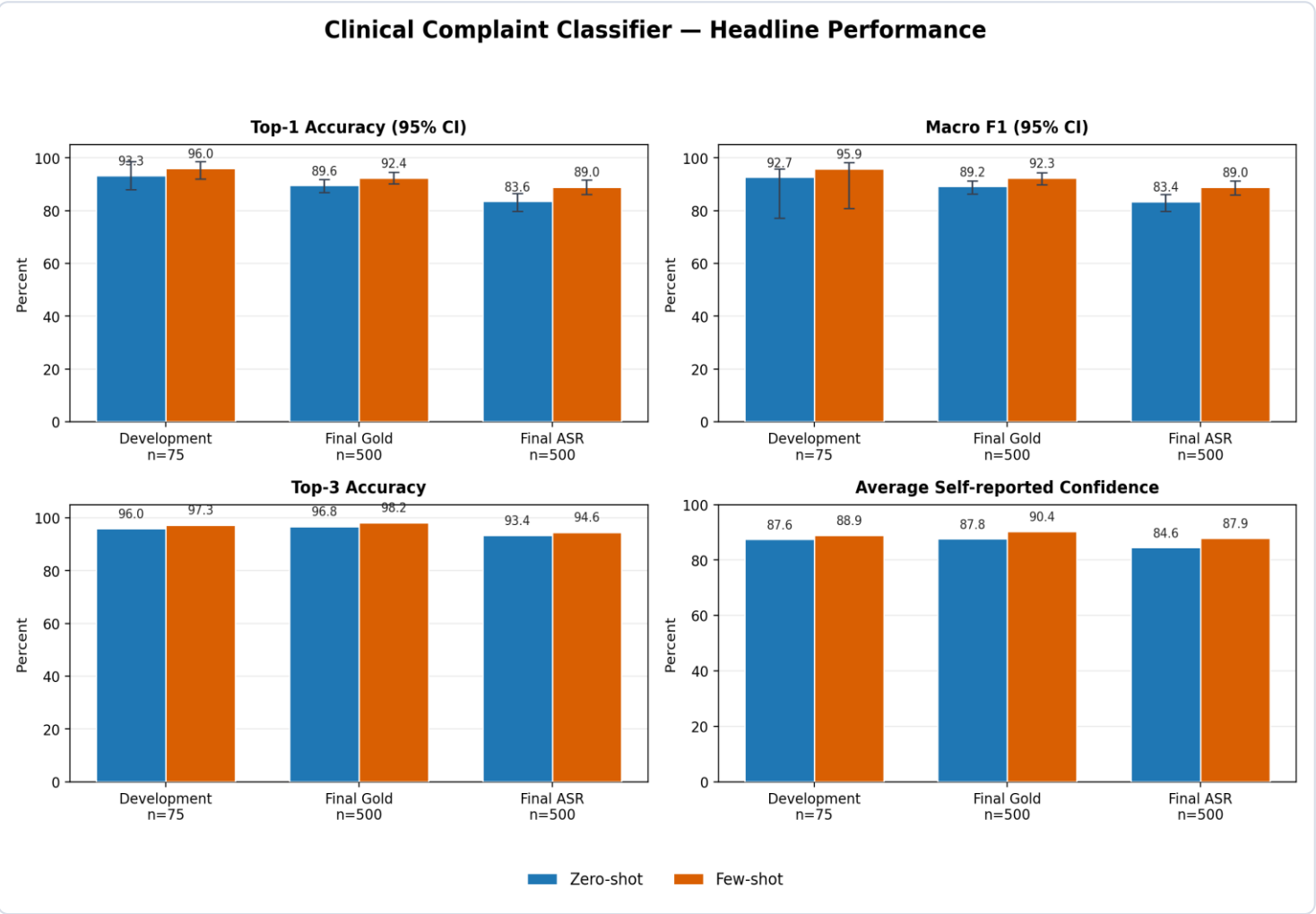
high-risk threshold

Higher-risk labels: Blurry vision, Hard to breath, Heart hurts, Infected wound, Internal pain

Review thresholds turn model confidence into a workflow decision: auto-accept, or send to a human.

Model Comparison Across Stages

Few-shot performs better than zero-shot, with the largest gain in the ASR setting.



KEY RESULTS

92.4%
few-shot final gold accuracy

89.6%
zero-shot final gold accuracy

89.0%
few-shot final ASR accuracy

83.6%
zero-shot final ASR accuracy

Final Gold Holdout Results

On clean reference transcripts, few-shot improves accuracy, macro-F1, and top-3 performance.

FINAL GOLD TEST SET

500

test examples

92.4%

few-shot accuracy
95% CI: 90.2–94.6%

92.3%

few-shot macro-F1
95% CI: 89.9–94.4%

98.2%

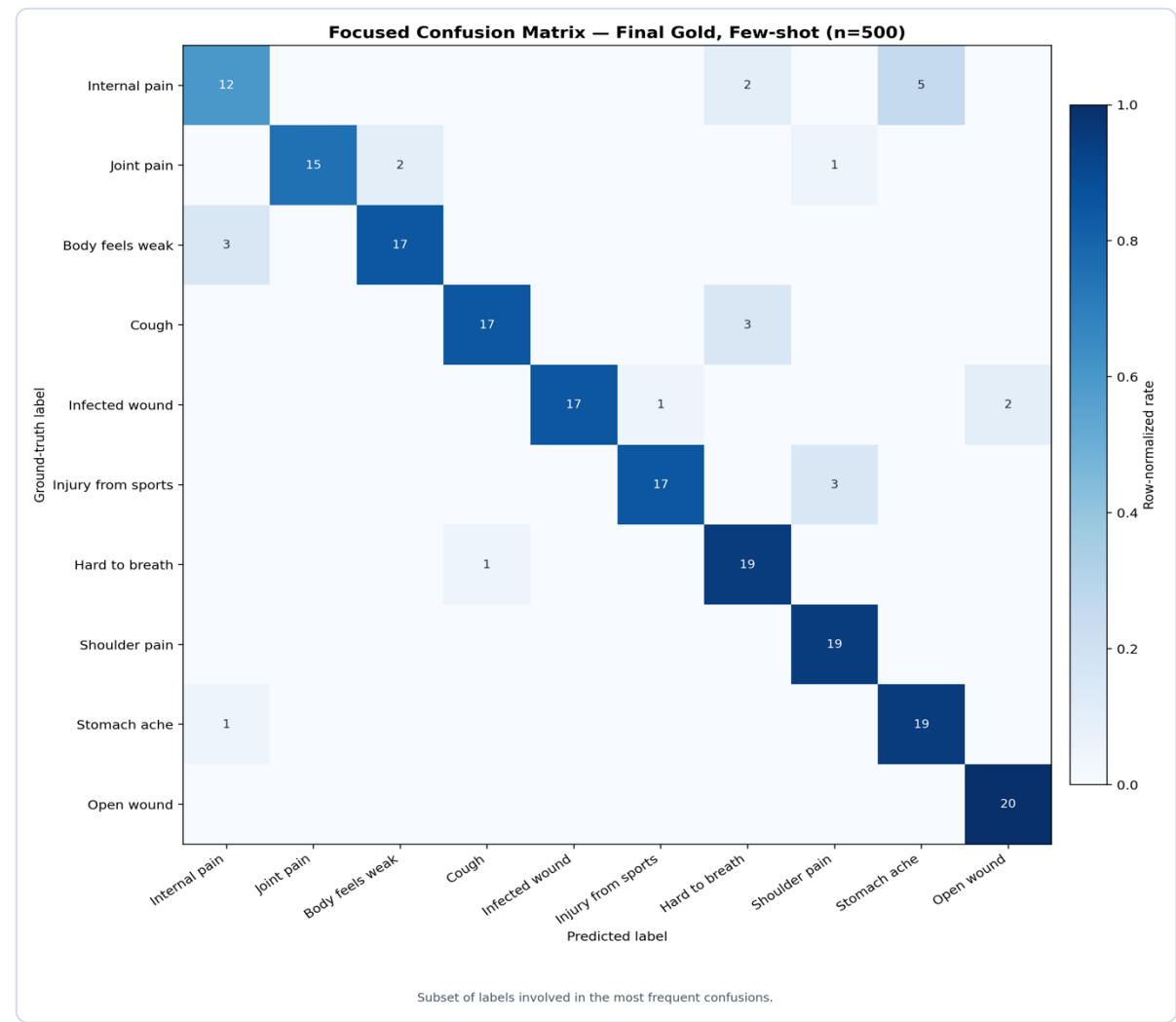
few-shot top-3 accuracy

INTERPRETATION

- Few-shot gains over zero-shot on clean transcripts: +2.8 percentage points in accuracy.
- Top-3 accuracy near 98% suggests the model often includes the correct label even when the top label is wrong.
- The clean-transcript result isolates classifier behavior before adding ASR noise.
- The remaining errors are important because they reveal label-boundary problems, not just random failures.

Error Analysis: Final Gold Confusion Matrix

Most clean-transcript errors occur between clinically adjacent or linguistically overlapping labels.



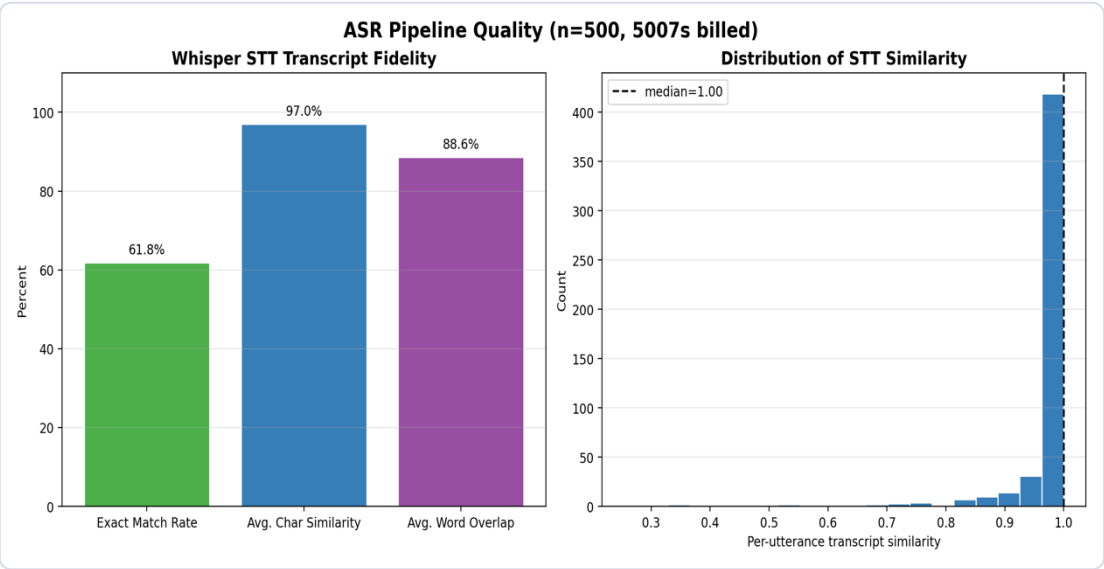
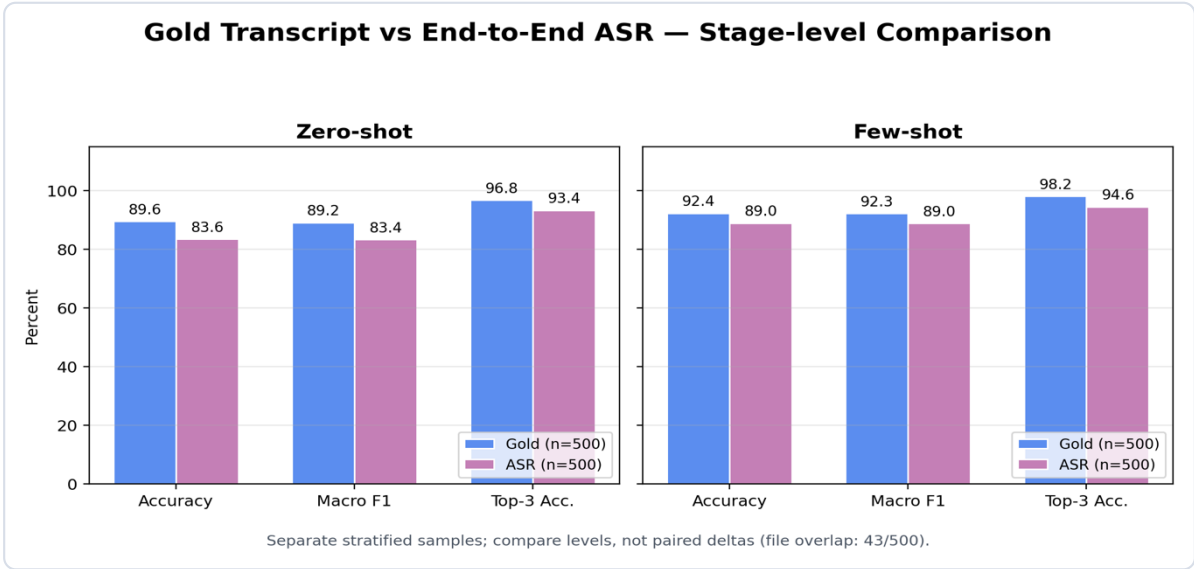
WHAT TO NOTICE

- Internal pain is the weakest final-gold few-shot label: F1 = 0.649.
- Most common confusion: Internal pain → Stomach ache, with 5 cases.
- Musculoskeletal labels also show boundary overlap, such as Muscle pain, Shoulder pain, and Joint pain.
- These are clinically interpretable errors, not random label noise.

Next-step implication: improve label definitions and add targeted examples for boundary cases.

Gold vs ASR Performance

The audio-to-label workflow remains strong, but ASR introduces a measurable performance drop.



92.4% → 89.0%

few-shot accuracy drop
Gold to ASR: -3.4 pp

98.2% → 94.6%

few-shot top-3 drop
Gold to ASR: -3.6 pp

0.970

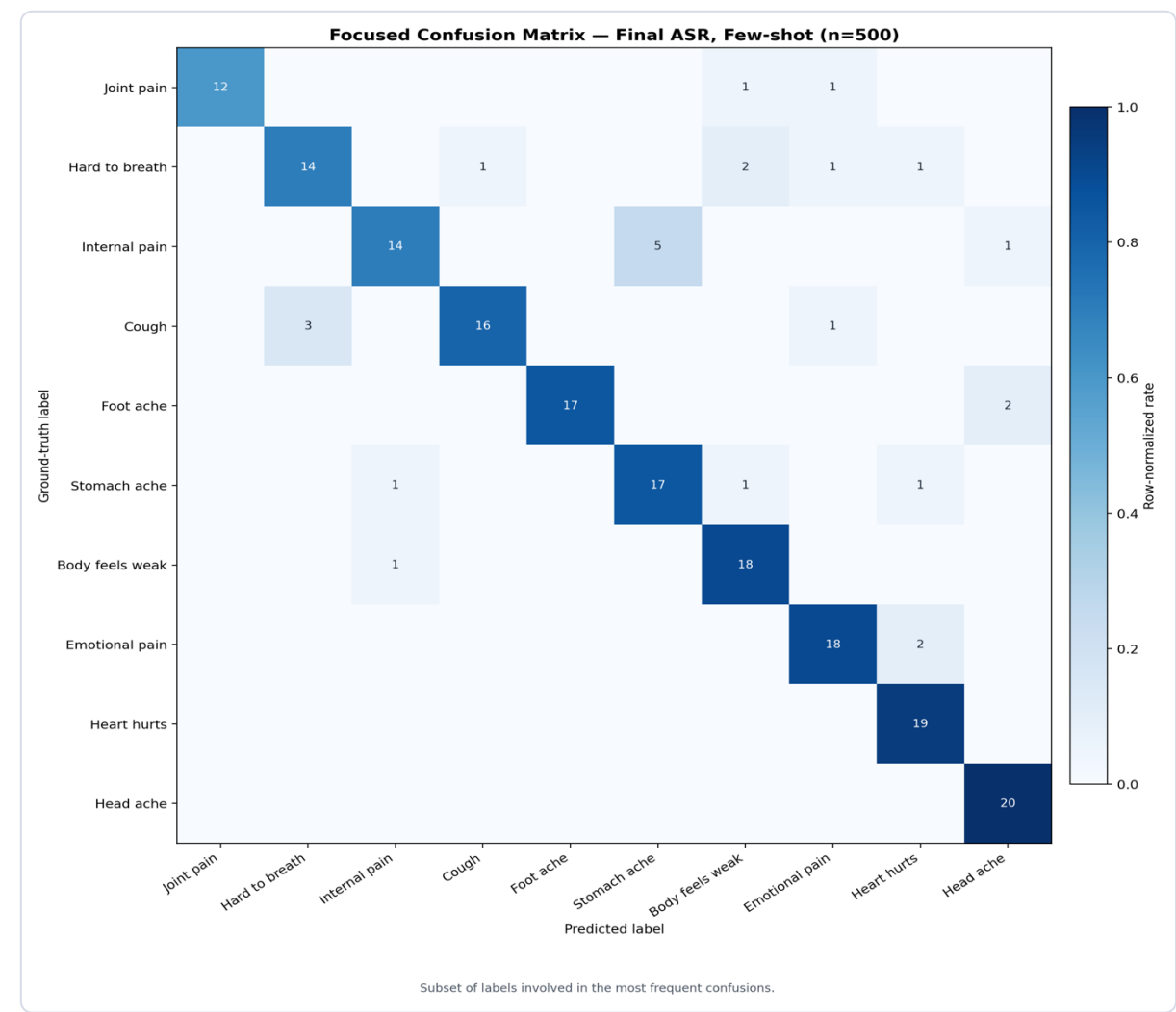
average transcript similarity

61.8%

exact transcript match rate

Error Analysis: Final ASR Confusion Matrix

The ASR setting changes not only accuracy, but also the pattern of mistakes.



WHAT THIS ADDS

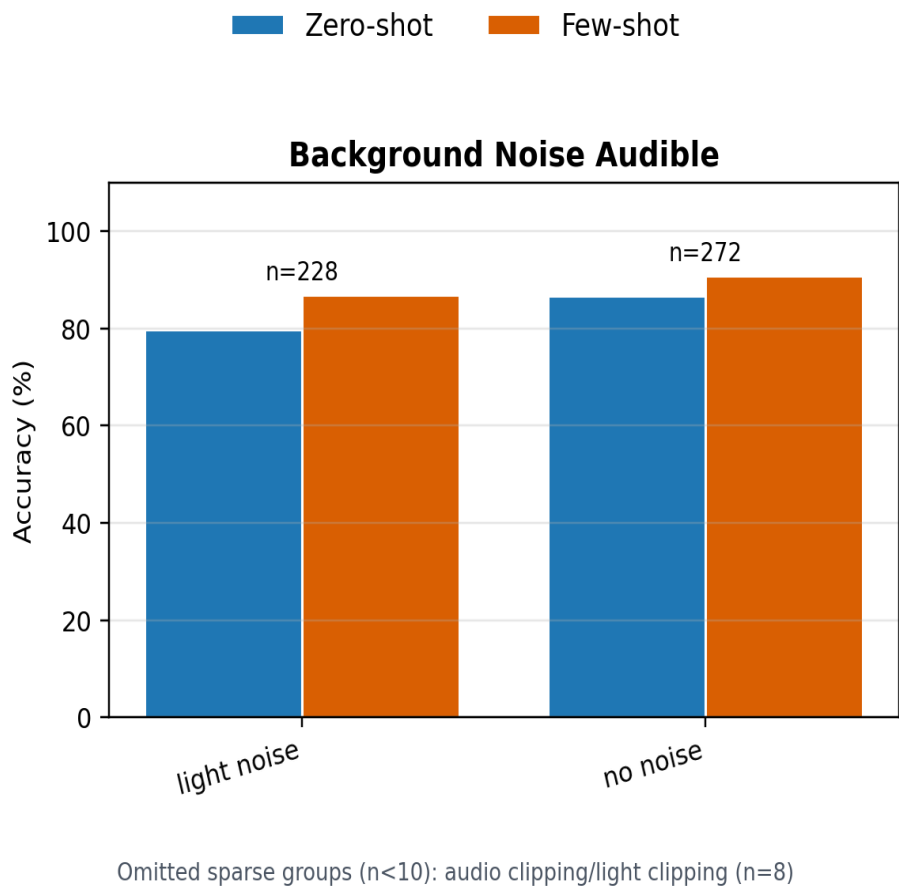
- This matrix asks whether audio transcription changes the error pattern.
- Final ASR few-shot accuracy is 89.0%, with macro-F1 = 0.890.
- Worst few-shot ASR labels include Internal pain, Joint pain, Hard to breath, Body feels weak, and Stomach ache.
- Errors still concentrate around overlapping symptom descriptions and anatomical labels.

Technical read: ASR noise can push borderline phrases across label boundaries, even when transcript similarity is high.

Audio Quality Subgroups

Background-noise findings are supported; clipping and quiet-speaker rows should be treated cautiously.

Subgroup Performance — Final ASR (n=500)



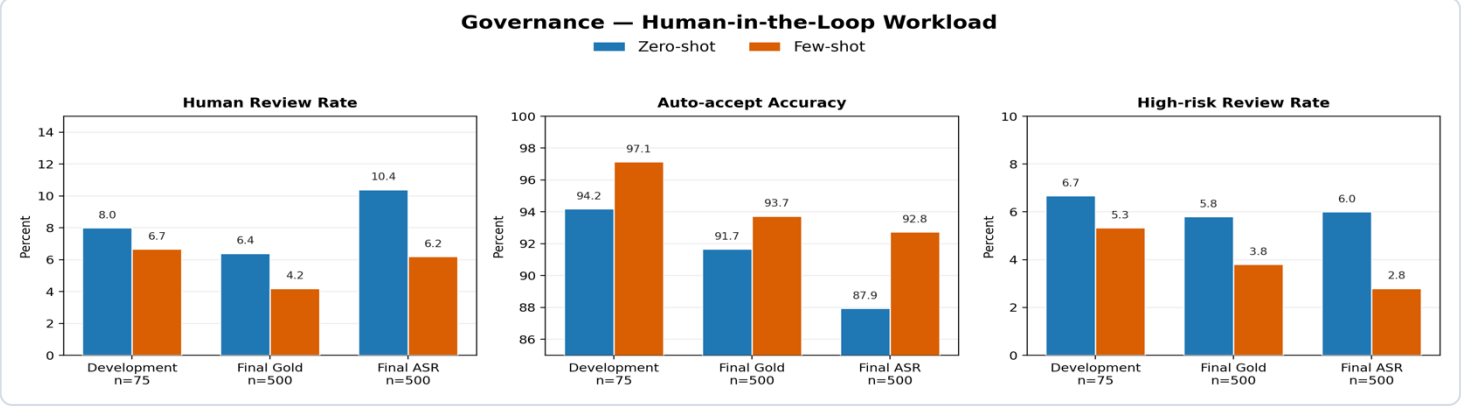
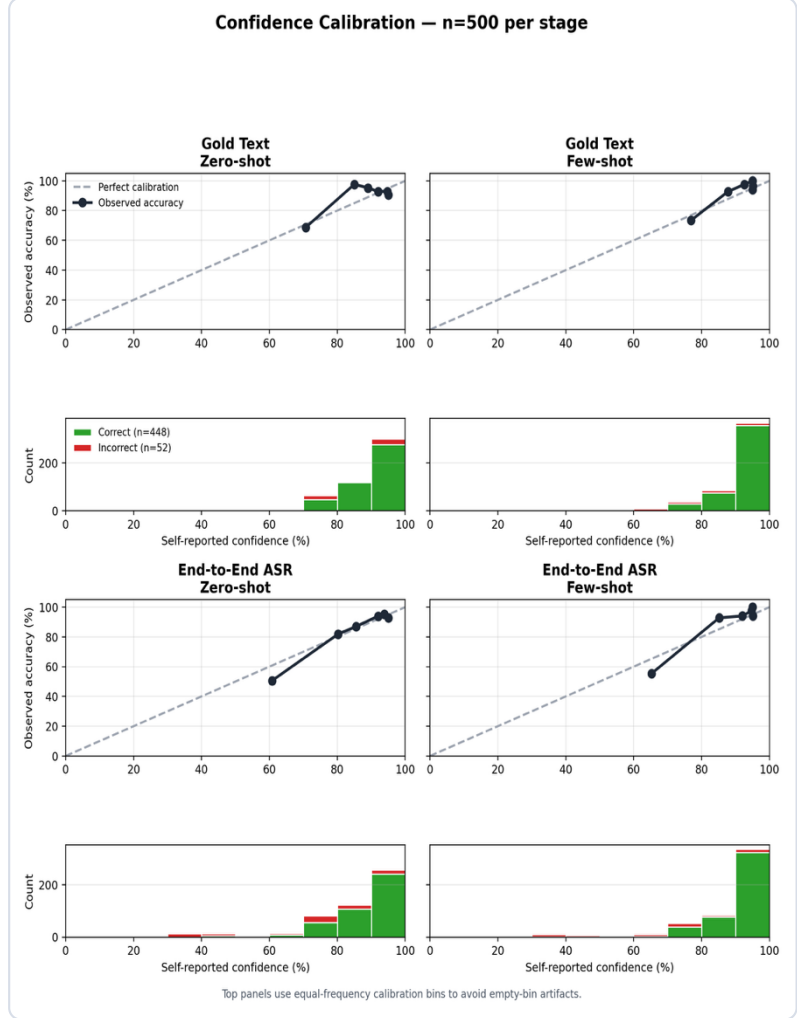
FEW-SHOT ASR SUBGROUP READ

Audio factor	Group	n	Accuracy	Interpretation
Noise	no_noise	272	90.8%	supported
Noise	light_noise	228	86.8%	supported
Clipping	light_clipping	8	75.0%	low support
Quiet speaker	audible_speaker	500	89.0%	no contrast

Submission-safe wording: clipping is hypothesis-generating only, and quiet-speaker effects cannot be evaluated in this run because there is no contrast group.

Confidence Calibration & Review Economics

Calibration is promising, but confidence should still be monitored before deployment.



CALIBRATION METRICS

Stage	Method	ECE-10	Brier
Final Gold	Few-shot	0.0396	0.0590
Final Gold	Zero-shot	0.0394	0.0832
Final ASR	Few-shot	0.0482	0.0692
Final ASR	Zero-shot	0.0334	0.1069

OPERATIONAL READ

- Few-shot final gold: 4.2% review rate; 93.7% auto-accept accuracy.
- Few-shot final ASR: 6.2% review rate; 92.8% auto-accept accuracy.
- Brier scores favor few-shot in both final stages.

Example Walkthrough

A single ASR case shows how retrieved examples can shift the model toward the intended label.

TRANSCRIPT

“I have a paint in my traps.”

Ground truth label: Muscle pain

ZERO-SHOT

Predicted: Shoulder pain
Confidence: 0.75
Top-3 includes Muscle pain

FEW-SHOT

Predicted: Muscle pain
Confidence: 0.92
Retrieved examples support symptom label

What this demonstrates: few-shot examples can help the model choose a symptom label instead of an adjacent anatomy/location label.

Parallel Baseline: Local Whisper + Classical Classifier

A second workflow gives us an offline/classical benchmark, but it is not a strict head-to-head with the LLM pipeline.

BEST LOCAL BASELINE

5,895

test examples in local workflow

84.5%

accuracy, best local model

0.839

macro-F1, best local model

94.1%

top-3 accuracy

Pipeline / model	Test n	Accuracy	Macro-F1	Top-3	How to interpret
Current few-shot LLM + API Whisper	500	89.0%	0.8897	94.6%	Higher exact-label performance
Local best baseline	5,895	84.5%	0.8387	94.1%	Offline/classical baseline; similar top-3
Local TF-IDF baseline	5,895	82.5%	0.8221	90.9%	Simpler text-feature model

Important caveat: this is a parallel benchmark, not an apples-to-apples comparison. The workflows differ in test size, classifier type, and transcript source.

- The local/classical baseline is useful because it shows what a non-LLM transcript classifier can do at larger scale.
- The LLM few-shot workflow remains stronger for top-1 label selection, while the local model is close in top-3 accuracy.
- This supports a practical tradeoff discussion: exact-label performance and flexibility versus offline execution and reduced API dependence.

API vs Local Whisper: Transcript Fidelity

Local Whisper reduces API dependence, but this run produced less exact transcripts and more missing outputs.

ASR COMPARISON

Metric	API Whisper workflow	Local Whisper cache	Interpretation
Evaluated examples	500	5,895 test entries	Different sample sizes
Average transcript similarity	0.9698	0.9169	Both high, API run cleaner
Exact transcript match	61.8%	14.1%	Local transcripts less verbatim
Average word overlap	0.8856	0.7839	Local run loses more wording detail
Empty transcripts	0 in ASR summary	474 test-cache entries	Local workflow needs fallback handling

Operational read: local ASR may be attractive for privacy, cost, and offline use, but this workflow needs stronger audio-quality checks and empty-transcript fallbacks.

DEFENSIBLE CONCLUSION

- API Whisper produced cleaner transcripts in the current evaluated ASR stage.
- Local Whisper still preserved enough meaning for a reasonable classical baseline, but transcript variability was higher.
- Before claiming local deployment readiness, rerun the same LLM prompts on the same examples using local Whisper transcripts.

Limitations & Future Directions

The system is promising, but deployment requires validation, calibration, and stress testing.

LIMITATIONS

- Internal dataset only; external validation is still required.
- Some labels have ambiguous clinical boundaries, especially Internal pain and Stomach ache.
- ASR subgroup analyses are unevenly powered; clipping has n=8 and quiet-speaker has no contrast group.
- Confidence calibration is useful but not yet deployment-proof.
- The workflow completed cleanly here, but production systems still need fallback handling.

FUTURE DIRECTIONS

- Validate on larger and more diverse clinical complaint datasets.
- Refine label definitions and add targeted boundary-case examples.
- Stress-test noisy, clipped, quiet, accented, and multilingual audio.
- Tune review thresholds prospectively with clinician input.
- Track malformed outputs, blocked responses, cost, latency, and audit logs in deployment.

Conclusions

Few-shot retrieval is the stronger prompting strategy, but the safest framing is supervised decision support.

MAIN CONCLUSIONS

- Few-shot improved final gold accuracy from 89.6% to 92.4%.
- Few-shot improved final ASR accuracy from 83.6% to 89.0%.
- Top-3 performance supports human-review workflows, especially when the top label is uncertain.
- Confidence thresholds reduce review workload, but calibration should be monitored.
- The local/classical workflow provides a useful offline baseline but should be treated as a parallel benchmark, not a strict replacement.

Takeaway: retrieved examples provide a practical upgrade for LLM complaint classification, while human review and validation remain essential for safe deployment.